

UNIVERSITY OF BIRMINGHAM

School of Mathematics

Programmes in the School of Mathematics

Programmes involving Mathematics

Second Examination

Second Examination

2LA 06 15552 Level I

Linear Algebra

May/June Examinations 2023-24

One Hour and Thirty Minutes

Full marks will be obtained with complete answers to BOTH questions. Each question carries equal weight. You are advised to initially spend no more than 45 minutes on each question and then to return to any incomplete questions if you have time at the end. An indication of the number of marks allocated to parts of questions is shown in square brackets.

No calculator is permitted in this examination.

Section A

1. (a) (i) Give the definition of a linear map.
- (ii) Verify that $\phi : P_1 \rightarrow P_1$ given by $f \mapsto \frac{d}{dx}[(x-1)f]$ is a linear map. Here P_1 stands for the space of all polynomials in the indeterminate x , of degree up to 1. [5]
- (b) Verify that the matrix of this ϕ with respect to the standard basis of P_1 is $M = \begin{pmatrix} 1 & -1 \\ 0 & 2 \end{pmatrix}$. [3]
- (c) (i) State the definition of the kernel of a linear map.
- (ii) Determine the kernel of the map ϕ from part (a).
- (ii) Is ϕ injective? Explain. [7]
- (d) Suppose that $\alpha : U \rightarrow V$ and $\beta : U \rightarrow V$ are linear maps, where U and V are vector spaces over the same field of scalars $\mathbb{F}$. Show that $\ker(\alpha) \subseteq \ker(\beta)$ if and only if, for $u, u' \in U$, the equality $\alpha(u) = \alpha(u')$ implies $\beta(u) = \beta(u')$. [5]
- (e) (i) Find the eigenvalues and eigenvectors of the matrix M above.
- (ii) Either find a matrix T such that $D = T^{-1}MT$ is diagonal or explain why M is not diagonalisable. [5]

Section B

2. (a) What is an inner product space? Your answer should include the definition of an inner product, but you may assume that the general concept of a bilinear form is known. [4]

- (b) Let V be an inner product space. For a subspace $W \subseteq V$, define the *perp* of W as

$$W^\perp = \{v \in V \mid (v, w) = 0 \text{ for all } w \in W\}.$$

Using the subspace criterion, show that $W^\perp$ is a subspace of V . [5]

- (c) Let again V be an inner product space.

(i) Which bases in V are called orthogonal? Which bases are called orthonormal?

(ii) Suppose that $B = \{v_1, v_2, \dots, v_n\}$ is an orthogonal basis of V . Deduce the following formula for the B -coordinates $c_1, c_2, \dots, c_n$ of a vector $v \in V$:

$$c_i = \frac{(v, v_i)}{(v_i, v_i)},$$

for $i = 1, 2, \dots, n$.

(iii) How does the formula simplify when the basis B is known to be orthonormal? [6]

- (d) Consider the polynomial space P_2 as an inner product space with respect to the inner product

$$(f, g) = \int_0^1 f(x)g(x)dx.$$

(i) Verify that the polynomials 1 and $2x - 1$ are orthogonal.

(ii) Find the length of $2x - 1$.

(iii) Find the orthogonal projection of x^2 to the line $L = \langle 1 \rangle$. Clearly state which polynomial is the projection. [5]

- (e) Consider $\mathbb{R}^4$ as an inner product space with respect to the dot product

$$u \cdot v = u_1v_1 + u_2v_2 + u_3v_3 + u_4v_4,$$

where $u = (u_1, u_2, u_3, u_4)$ and $v = (v_1, v_2, v_3, v_4)$. Use the Gram-Schmidt process to construct an orthogonal basis in the subspace $E = \langle (1, -1, 0, 0), (1, -1, -1, 1), (2, 0, 0, 2) \rangle$. [5]

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LI Linear Algebra

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats and outer-wear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) **MUST** be placed in the designated area.
- Check that you **DO NOT** have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **MUST** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are **NOT** permitted to use a mobile phone as a clock. If you have difficulty in seeing a clock, please alert an Invigilator.
- You are **NOT** permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately.
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to the Student Conduct procedures.